

Air University



Network Security (Lab)

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Department of Cyber Security

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Lab # 15

Task 1 - Substitution Cipher

i. Python Key Generator

```
(kali㉿kali)-[~/Documents/Lab_15]
$ nano cipher.py

(kali㉿kali)-[~/Documents/Lab_15]
$ python3 cipher.py
Generated Substitution Cipher Key:
4u21wt1sudvxaz7ahasrmsrya4
```

Command breakdown:

- `random.sample(S, len(S))` — picks all 26 letters in random order without repetition [Certain]
- `"".join(list)` — joins the list into a single string

ii. Create article.txt

```
(kali㉿kali)-[~/Documents/Lab_15]
$ nano article.txt

(kali㉿kali)-[~/Documents/Lab_15]
$ cat article.txt
Lion and the mouse is a story that beautifully illustrates the importance of showing kindness and gratitude. It reminds us that true strength comes from humility and that compassion knows no boundaries. The story emphasizes the significance of helping others, no matter how insignificant they may seem. Any act of kindness, whether big or small, can have a profound impact on the lives of others. The lion and the mouse showcase that even the most unlikely of friendships can blossom from a simple act of compassion."
```

iii. Convert to lowercase

```
(kali㉿kali)-[~/Documents/Lab_15]
$ tr '[:upper:]' '[:lower:]' < article.txt > lowercase.txt

(kali㉿kali)-[~/Documents/Lab_15]
$ ls
article.txt  cipher.py  lowercase.txt
```

Command breakdown:

- `tr` — transliterates (replaces) characters
- `'[:upper:]'` — matches any uppercase letter
- `'[:lower:]'` — replaces with the corresponding lowercase letter
- `< article.txt` — feeds the file as input
- `> lowercase.txt` — redirects output to a new file

iv. Strip punctuation/numbers to make plaintext

```
(kali@kali)-[~/Documents/Lab_15]
$ tr -cd '[a-z][\n][:space:]' < lowercase.txt > plaintext.txt

(kali@kali)-[~/Documents/Lab_15]
$ sdiff lowercase.txt plaintext.txt
lion and the mouse is a story that beautifully illustrates th | lion and the mouse is a story that beautifully illustrates th
kindness and gratitude. it reminds us that true strength come | kindness and gratitude it reminds us that true strength comes
compassion knows no boundaries. the story emphasizes the sign | compassion knows no boundaries the story emphasizes the signi
no matter how insignificant they may seem. | no matter how insignificant they may seem
any act of kindness, whether big or small, can have a profound | any act of kindness whether big or small can have a profound
the lion and the mouse showcase that even the most unlikely o | the lion and the mouse showcase that even the most unlikely o
from a simple act of compassion." | from a simple act of compassion

(kali@kali)-[~/Documents/Lab_15]
$ tr 'abcdefghijklmnopqrstuvwxyz' 'tmkadlnfwogyqrbcuijzhvpe' < plaintext.txt > ciphertext.txt
```

Command breakdown:

- -c — complement (inverts the character set — keeps everything NOT listed)
- -d — delete characters in the set
- [a-z][\n][:space:] — the characters to keep: lowercase letters, newlines, and spaces

v. Encrypt plaintext to ciphertext using tr

```
(kali@kali)-[~/Documents/Lab_15]
$ tr 'abcdefghijklmnopqrstuvwxyz' 'tmkadlnfwogyqrbcuijzhvpe' < plaintext.txt > ciphertext.txt

(kali@kali)-[~/Documents/Lab_15]
$ cat ciphertext.txt
gfrq tqa ind yrjud fu t uirse inti mdtjifljgge fggjuistidu ind fybrsitqkd rl unrhfqx
ofqaqduu tqa xstifijad fi sdyfqau ju inti isjd uisdqxin krydu lsry njyfgfie tqa inti
krybtuufqr oqrhu qr mrjqatsfdu ind uirse dybntufpdu ind ufxqflfktqkd rl ndgbfqx rindsu
qr ytiids nrh fqufxqflfktqi inde yte uddy
tqe tki rl ofqaqduu hndinds mfx rs uytgg ktq ntzd t bsrlrjqqa fybtqi rq ind gfzdu rl rindsu
ind gfrq tqa ind yrjud unrhktud inti dzdq ind yrui jggfodge rl lsfdqaunfbu ktq mgruury
lsry t ufbygd tki rl krybtuufqr
```

Command breakdown:

- First string 'usamaarshad247141rstuvwxyz' — the original alphabet (what to find)
- Second string 'vudgbksorlmqhxjctenwizpfya' — the substitution key (what to replace with)
- tr maps each character positionally: a→v, b→u, c→d, etc.

vi. Decrypt ciphertext back to plaintext

```
(kali@kali)-[~/Documents/Lab_15]
$ tr 'tmkadlnfwogyqrbcuijzhvpe' 'abcdefghijklmnopqrstuvwxyz' < ciphertext.txt > decrypted.txt

(kali@kali)-[~/Documents/Lab_15]
$ cat decrypted.txt
lion and the mouse is a story that beautifully illustrates the importance of showing
kindness and gratitude it reminds us that true strength comes from humility and that
compassion knows no boundaries the story emphasizes the significance of helping others
no matter how insignificant they may seem
any act of kindness whether big or small can have a profound impact on the lives of others
the lion and the mouse showcase that even the most unlikely of friendships can blossom
from a simple act of compassion

(kali@kali)-[~/Documents/Lab_15]
$ sdiff decrypted.txt plaintext.txt
lion and the mouse is a story that beautifully illustrates th | lion and the mouse is a story that beautifully illustrates th
kindness and gratitude it reminds us that true strength comes | kindness and gratitude it reminds us that true strength comes
compassion knows no boundaries the story emphasizes the signi | compassion knows no boundaries the story emphasizes the signi
no matter how insignificant they may seem | no matter how insignificant they may seem
any act of kindness whether big or small can have a profound | any act of kindness whether big or small can have a profound
the lion and the mouse showcase that even the most unlikely o | the lion and the mouse showcase that even the most unlikely o
from a simple act of compassion | from a simple act of compassion
```

Task 2 - ECB and CBC Image Encryption with OpenSSL

i. Get a BMP image

```
(kali㉿kali)-[~/Documents/Lab_15]
$ convert /home/kali/Desktop/download.jpeg -compress None pic_original.bmp

(kali㉿kali)-[~/Documents/Lab_15]
$ file pic_original.bmp
pic_original.bmp: PC bitmap, Windows 98/2000 and newer format, 736 x 397 x 24, cbSize 876714, bits offset 138

(kali㉿kali)-[~/Documents/Lab_15]
$ ls -lh pic_original.bmp
-rw-rw-r-- 1 kali kali 857K May 20 05:36 pic_original.bmp
```

ii. Encrypt with ECB mode

```
(kali㉿kali)-[~/Documents/Lab_15]
$ openssl enc -aes-128-ecb -e \
  -in pic_original.bmp \
  -out cipher_ecb.bmp \
  -K 00112233445566778899aabbccddeeff \
  -nosalt
```

Command breakdown:

- openssl enc — invokes the symmetric encryption tool
- -aes-128-ecb — AES algorithm, 128-bit key, ECB mode
- -e — encrypt (default, but explicit for clarity)
- -in pic_original.bmp — input file
- -out cipher_ecb.bmp — output file
- -K 00112233445566778899aabbccddeeff — the 128-bit key in hex (32 hex chars = 16 bytes = 128 bits)
- -nosalt — disables the random salt prefix; needed here so the raw encrypted bytes start at byte 0 and we can do clean header manipulation

iii. Reconstruct viewable ECB-encrypted image

```
(kali㉿kali)-[~/Documents/Lab_15]
$ head -c 54 pic_original.bmp > header

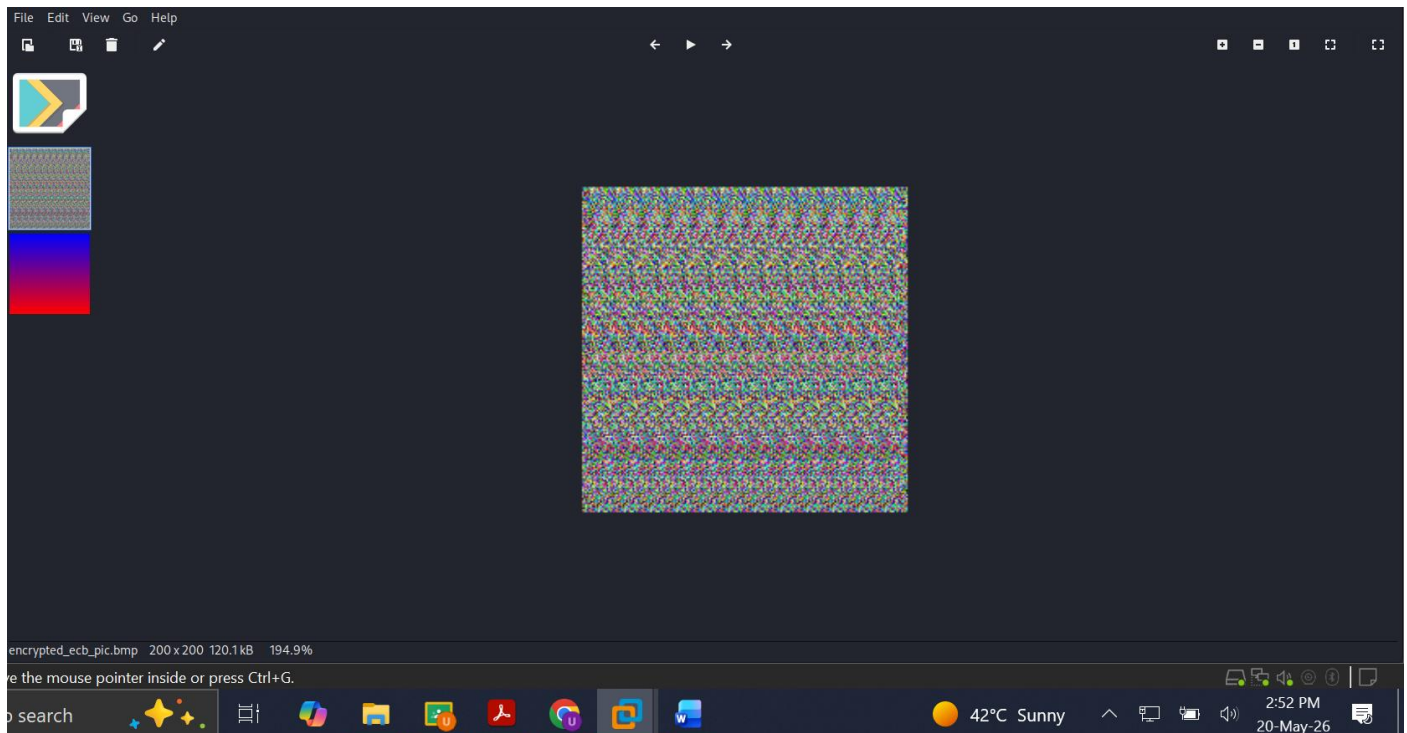
(kali㉿kali)-[~/Documents/Lab_15]
$ tail -c +55 cipher_ecb.bmp > body

(kali㉿kali)-[~/Documents/Lab_15]
$ cat header body > encrypted_ecb_pic.bmp
```

Command breakdown:

- head -c 54 — outputs the first 54 bytes (the BMP header) [Certain]
- tail -c +55 — outputs everything starting from byte 55 onward (skips the header) [Certain]
- cat header body > encrypted_ecb_pic.bmp — concatenates the two files

```
(kali㉿kali)-[~/Documents/Lab_15]
$ xdg-open encrypted_ecb_pic.bmp
```



iv. Encrypt with CBC mode

```
(kali㉿kali)-[~/Documents/Lab_15]
$ openssl enc -aes-128-cbc -e \
  -in pic_original.bmp \
  -out cipher_cbc.bmp \
  -K 00112233445566778899aabbccddeeff \
  -iv 0102030405060708090a0b0c0d0e0f10 \
  -nosalt

(kali㉿kali)-[~/Documents/Lab_15]
$ head -c 54 pic_original.bmp > header

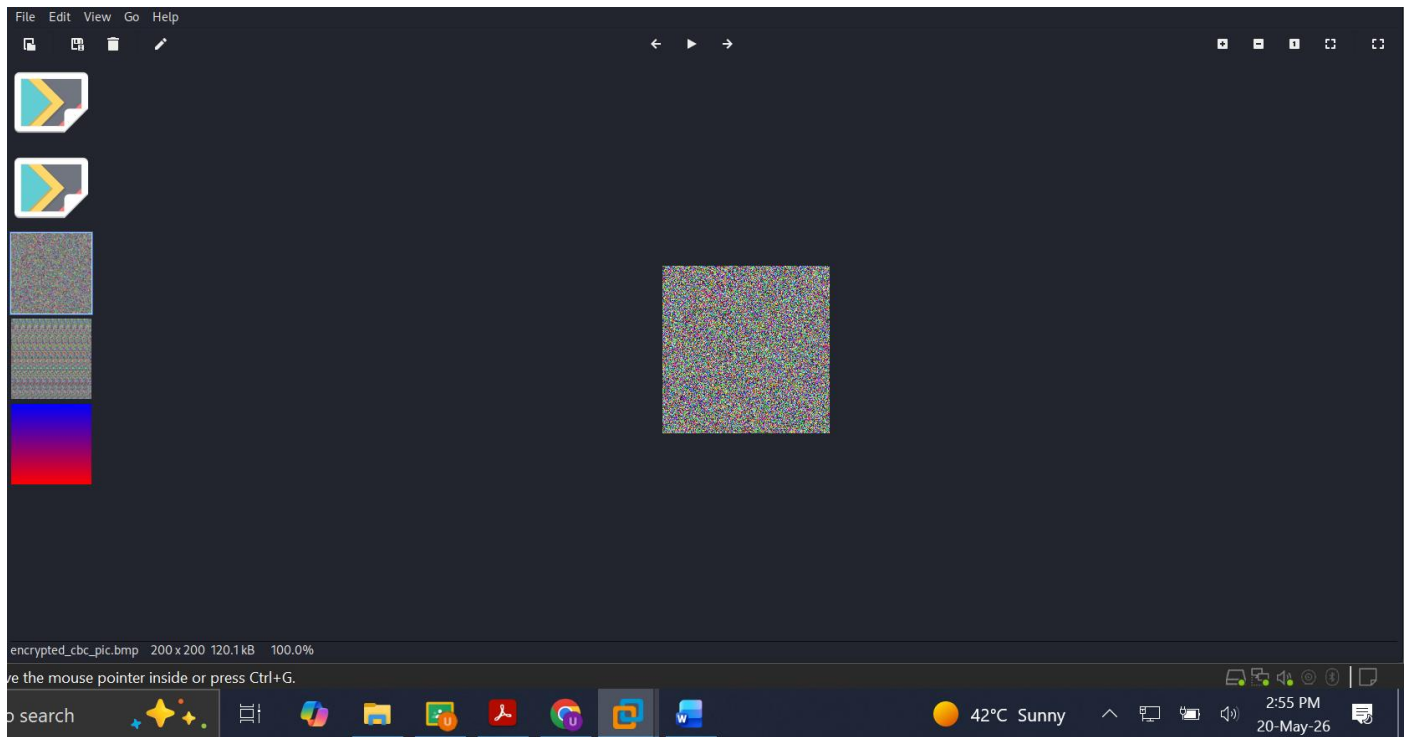
(kali㉿kali)-[~/Documents/Lab_15]
$ tail -c +55 cipher_cbc.bmp > body

(kali㉿kali)-[~/Documents/Lab_15]
$ cat header body > encrypted_cbc_pic.bmp

(kali㉿kali)-[~/Documents/Lab_15]
$ xdg-open encrypted_cbc_pic.bmp
```

Additional flag:

- `-iv 0102030405060708090a0b0c0d0e0f10` — the Initialization Vector (IV), 16 bytes (128-bit) in hex; required for CBC because the first block has no previous ciphertext to XOR with.



v. Key Differences

Feature	ECB	CBC
Pattern leakage	Yes — identical blocks produce identical ciphertext	No — chaining randomizes output
IV required	No	Yes
Parallel encryption	Possible	Not possible (each block depends on previous)
Security	Weak	Stronger

- End of Lab Task -